

Geometry

Unit 10

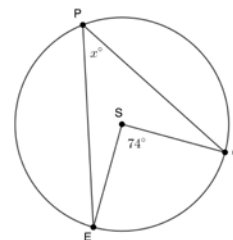
Lesson 6 Homework

Name _____

Date _____

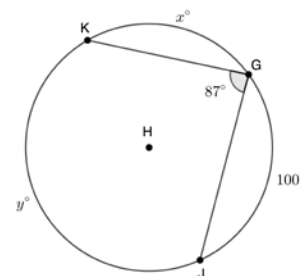
Period _____

- Circle S is shown where $\angle GSE$ is a central angle, $\angle GPE$ is inscribed in the circle, and $m\angle GSE = 74^\circ$. Find $m\angle GPE$.



For questions 2-3, Circle H has points K , G , and J on the circle. $\angle KGJ$ is inscribed in Circle H measuring 87° , and the $m\widehat{GJ} = 100^\circ$.

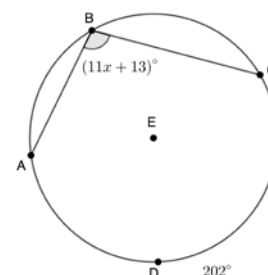
- Find the $m\widehat{KJ}$.



- Find the $m\widehat{KG}$.

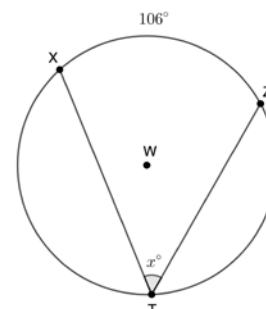
For questions 4-5, Points A , B , C , and D lie on Circle E . $\angle ABC$ is inscribed in the circle, the $m\angle ABC = (11x + 13)^\circ$, and the $m\widehat{CDA} = 202^\circ$.

- Find the $m\angle ABC$.



- Find the value of x .

- Circle W is shown, where points X , Z , and T are on the circle. $\angle XTZ$ is inscribed in Circle W with $m\widehat{XZ} = 106^\circ$. Find the value of x .



For questions 7-10, Circle X is shown with inscribed angle $\angle ABC$ and diameter $\overline{CA}$. The $m\widehat{AB} = (14x - 16)^\circ$ and the $m\widehat{BC} = (10x - 20)^\circ$.

7. Find the value of x .

8. Find the $m\angle BAC$.

9. Find the $m\angle ACB$.

10. Find the $m\widehat{CB}$.

